

Step 1: Title Page

- KhetHub Internship Program 2026
 - **Task 4** – Smart Agriculture product proposal challenges
 - Name:- Pranshu Tyagi
 - Role: Web Development Intern
 - Date:- 06/07/2026
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This task is evaluating your **product thinking + web development knowledge**, not just coding. Since you're applying as a **Web Development Intern**, you should explain the full architecture in a professional way.

KhetHub Internship Task 4 Proposal

Smart Water Management & Irrigation Advisor

1. Problem Identification

Problem

Many farmers do not know **when** and **how much** to irrigate their crops.

As a result:

- 💧 Water is wasted.
- 🌿 Crops receive too much or too little water.
- 📉 Crop yield decreases.
- 💰 Electricity and diesel costs increase.

Research

According to agricultural studies, irrigation decisions are often based on experience rather than real-time field conditions. This leads to inefficient water usage, especially during uncertain weather.

2. Proposed Solution

Product Name

Smart Irrigation Advisor

Feature Overview

A digital feature inside KhetHub that recommends the best irrigation schedule using:

- Weather forecast
- Soil moisture
- Crop type

- Crop growth stage
 - Farm location
-

Target Users

- Small farmers
 - Medium farmers
 - Large farms
 - Agricultural advisors
-

Key Functionalities

Weather Prediction

Shows rainfall prediction.

Irrigation Recommendation

Example:

Your wheat field requires irrigation tomorrow morning for 35 minutes.

Crop Selection

Farmer selects:

- Wheat
 - Rice
 - Potato
 - Tomato
 - etc.
-

GPS Location

Automatically detects village/location.

Notification

Send reminder:

Irrigate your field at 6:00 AM tomorrow.

Water Usage History

Displays:

- Total irrigation
- Water saved
- Previous recommendations

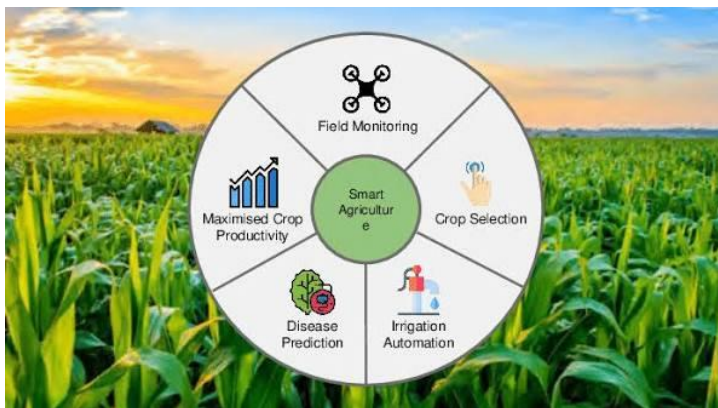
Benefits for Farmers

- ✓ Save Water
- ✓ Higher Yield
- ✓ Lower Electricity Cost
- ✓ Better Crop Health
- ✓ Scientific Decision Making

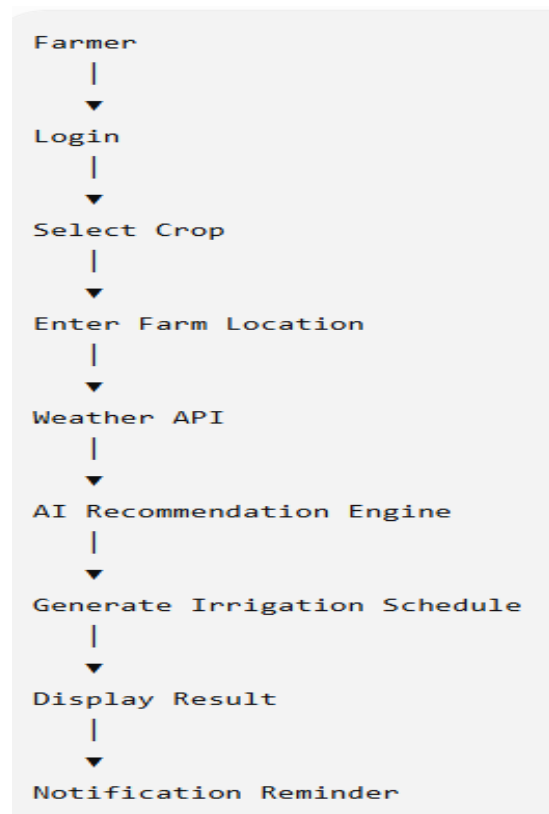
Why KhethHub Should Build It

This feature:

- Solves a real farming problem
- Can reach millions of farmers
- Improves customer engagement
- Supports sustainable farming
- Can later integrate with IoT devices



3. Product Workflow



Simple Wireframe

Smart Irrigation Advisor

Crop:
[Wheat ▼]

Location:
Meerut

Soil Moisture:
38%

Weather:
Rain Tomorrow

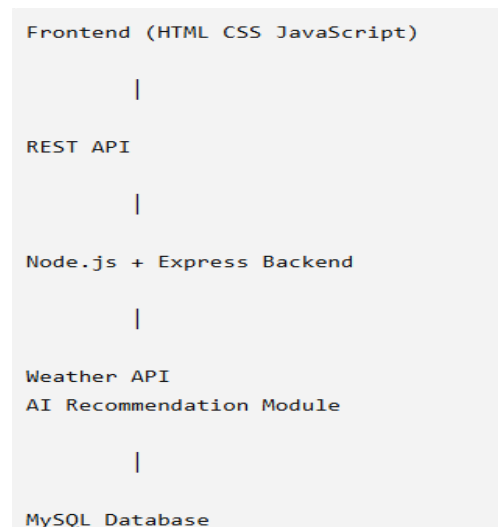
Recommendation

Irrigate Tomorrow
Duration:
35 Minutes

[Start Irrigation]

4. Technical Approach

System Architecture



Frontend

Technology:

- HTML
- CSS
- JavaScript

Pages:

- Login
- Dashboard
- Crop Selection
- Recommendation Screen
- History Page

Backend

Technology:

- Node.js
- Express.js

Responsibilities:

- User Authentication
 - Fetch Weather
 - Generate Recommendation
 - Store Farmer Data
 - Notification API
-

Database

Users Table

Field	Type
user_id	INT
name	VARCHAR
mobile	VARCHAR
village	VARCHAR

Farms

Field	Type
farm_id	INT
user_id	INT
crop	VARCHAR
area	FLOAT

Irrigation Records

Field	Type
record_id	INT
farm_id	INT
moisture	FLOAT
recommendation	TEXT
date	DATE

API Design

Login

```
POST /api/login
```

Request

```
</> JSON
{
  "mobile": "9876543210",
  "password": "123456"
}
```

Response

```
</> JSON
{
  "status": "success",
  "token": "JWT_TOKEN"
}
```

Get Recommendation

```
GET /api/recommendation
```

Response

«» JSON

```
{
  "crop": "Wheat",
  "soilMoisture": "38%",
  "weather": "Rain Tomorrow",
  "recommendation": "Irrigate Tomorrow Morning",
  "duration": "35 Minutes"
}
```

Save Recommendation

POST /api/history

Fetch History

GET /api/history

Backend Workflow

User Login

↓

Authentication

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Fetch Farm Details

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Call Weather API

↓

Analyze Soil Moisture

↓

Generate Recommendation

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Store Database

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Return JSON Response

↓

Display Dashboard

Bonus Features

AI Integration

AI predicts:

- Disease risk
 - Water requirement
 - Crop stress
-

Voice Assistant

Farmer asks:

"Kal paani dena hai?"

Assistant replies in Hindi.

Multilingual Support

- Hindi
 - English
 - Punjabi
 - Marathi
 - Gujarati
 - Tamil
-

IoT Integration

Connect:

- Soil Moisture Sensor
 - Smart Irrigation Pump
-

Offline Mode

Farmer can view the last downloaded recommendation without internet.

GPS Service

Automatically identifies:

- District

- Weather
- Nearby mandi

☑ Predictive Analytics

Shows:

- Expected water requirement
- Rainfall prediction
- Water-saving statistics

UI Suggestion

Dashboard Sections

1. 🌤 Today's Weather
2. 🌿 Crop Details
3. 💧 Irrigation Recommendation
4. 📊 Water Usage Report
5. 🔔 Notifications
6. 📄 History

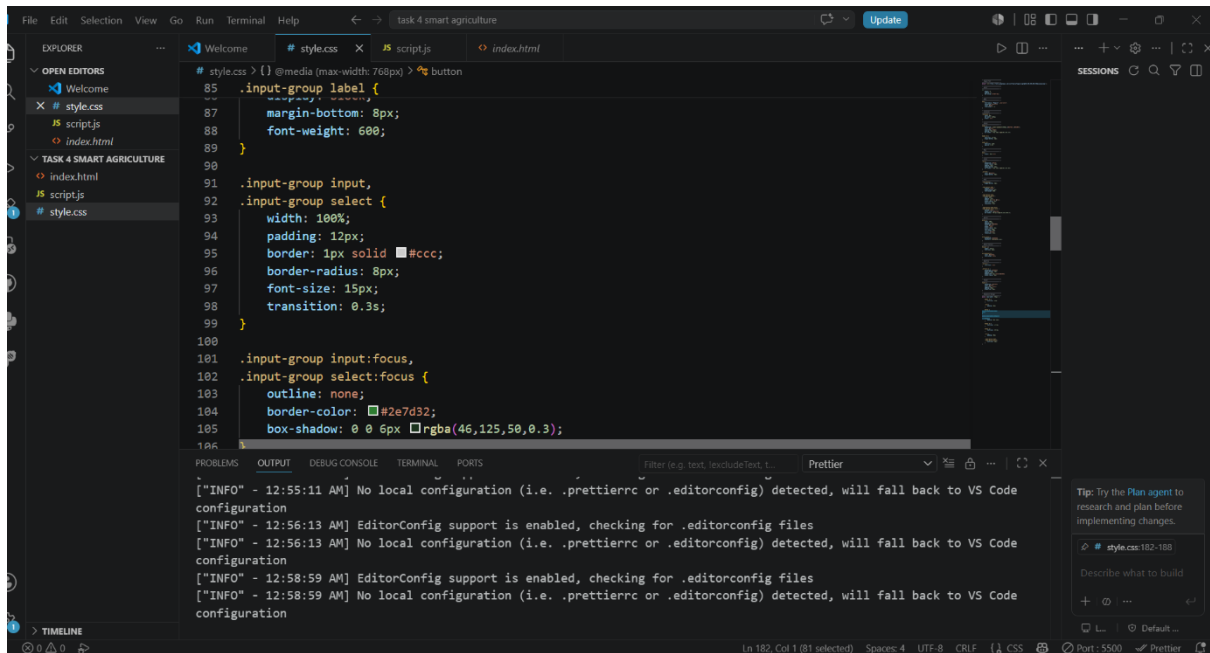
Index.html

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2  <html lang="en">
16 <body>
25 <main class="container">
27   <section class="card">
31     <form id="farmForm">
32       <input
36         type="text"
37         id="farmerName"
38         placeholder="Enter your name"
39         required
40       >
41     </div>
42
43     <div class="input-group">
44       <label for="crop">Select Crop</label>
45
46       <select id="crop" required>
47         <option value="">Choose Crop</option>
48         <option>Wheat</option>
49         <option>Rice</option>
50         <option>Potato</option>
51         <option>Tomato</option>
52         <option>Sugarcane</option>
53         <option>Maize</option>
54       </select>
55     </div>
56
57     <div class="input-group">
58       <label for="location">Location</label>
59       <input
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Style.css



```
# style.css > {} @media (max-width: 768px) > button
85 .input-group label {
86     margin-bottom: 8px;
87     font-weight: 600;
88 }
89
90
91 .input-group input,
92 .input-group select {
93     width: 100%;
94     padding: 12px;
95     border: 1px solid #ccc;
96     border-radius: 8px;
97     font-size: 15px;
98     transition: 0.3s;
99 }
100
101 .input-group input:focus,
102 .input-group select:focus {
103     outline: none;
104     border-color: #2e7d32;
105     box-shadow: 0 0 6px #4682b4;
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, excludeText, ...) Prettier

["INFO" - 12:55:11 AM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

["INFO" - 12:56:13 AM] EditorConfig support is enabled, checking for .editorconfig files

["INFO" - 12:56:13 AM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

["INFO" - 12:58:59 AM] EditorConfig support is enabled, checking for .editorconfig files

["INFO" - 12:58:59 AM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

Tip: Try the Plan agent to research and plan before implementing changes.

+ # style.css:182-188

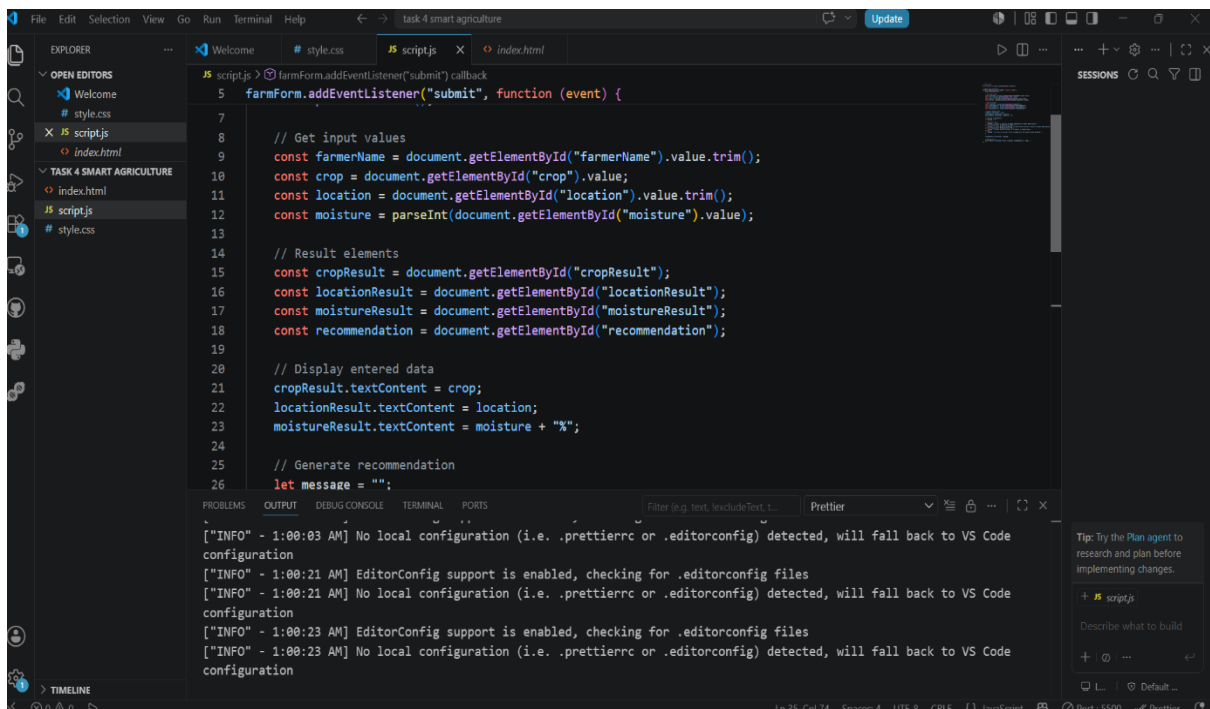
Describe what to build

+ @ ...

Default ...

Port: 5500 Prettier

Script.js



```
JS script.js > farmForm.addEventListener("submit") callback
5 farmForm.addEventListener("submit", function (event) {
6
7     // Get input values
8     const farmerName = document.getElementById("farmerName").value.trim();
9     const crop = document.getElementById("crop").value;
10    const location = document.getElementById("location").value.trim();
11    const moisture = parseInt(document.getElementById("moisture").value);
12
13    // Result elements
14    const cropResult = document.getElementById("cropResult");
15    const locationResult = document.getElementById("locationResult");
16    const moistureResult = document.getElementById("moistureResult");
17    const recommendation = document.getElementById("recommendation");
18
19    // Display entered data
20    cropResult.textContent = crop;
21    locationResult.textContent = location;
22    moistureResult.textContent = moisture + "%";
23
24    // Generate recommendation
25    let message = "";
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, excludeText, ...) Prettier

["INFO" - 1:00:03 AM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

["INFO" - 1:00:21 AM] EditorConfig support is enabled, checking for .editorconfig files

["INFO" - 1:00:21 AM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

["INFO" - 1:00:23 AM] EditorConfig support is enabled, checking for .editorconfig files

["INFO" - 1:00:23 AM] No local configuration (i.e. .prettierrc or .editorconfig) detected, will fall back to VS Code configuration

Tip: Try the Plan agent to research and plan before implementing changes.

+ JS script.js

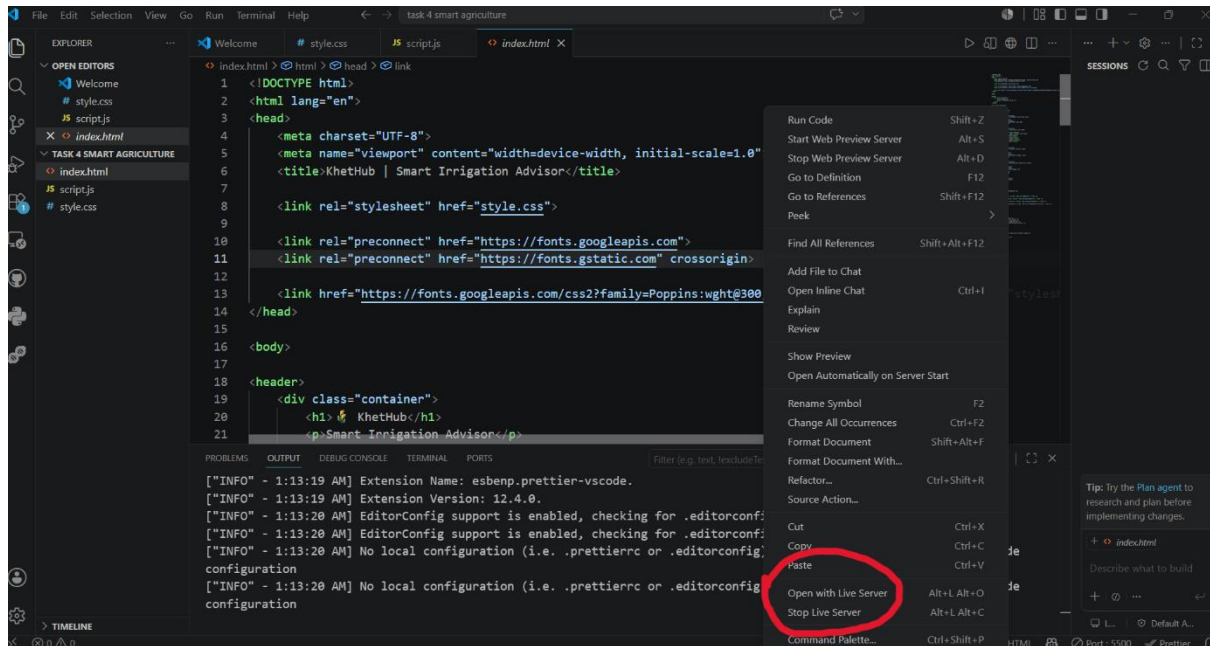
Describe what to build

+ @ ...

Default ...

Port: 5500 Prettier

Live server through index.html



Before



Smart Irrigation Advisor

Farm Details

Farmer Name

Select Crop

Location

Soil Moisture (%)

Irrigation Recommendation

Crop: --

Location: --

Soil Moisture: --

Recommendation: Waiting...

Benefits

-  Save Water
-  Improve Crop Health
-  Increase Crop Yield
-  Reduce Irrigation Cost
-  Smart Weather-Based Decisions

And after that

Smart Agriculture dashboard challenges

A screenshot of a web browser showing the KhetHub Smart Irrigation Advisor interface. The browser's address bar displays '127.0.0.1:5500/index.html'. The page has a green header with the KhetHub logo and the text 'Smart Irrigation Advisor'. Below the header is a white card titled 'Farm Details'. This card contains four input fields: 'Farmer Name' with the value 'Pranshu Tyagi', 'Select Crop' with a dropdown menu showing 'Rice', 'Location' with the value 'meerut', and 'Soil Moisture (%)' with the value '10'. At the bottom of the card is a green button labeled 'Get Recommendation'.

KhetHub
Smart Irrigation Advisor

Farm Details

Farmer Name
Pranshu Tyagi

Select Crop
Rice

Location
meerut

Soil Moisture (%)
10

Get Recommendation

A screenshot of the KhetHub Smart Irrigation Advisor results page. The browser's address bar shows '127.0.0.1:5500/index.html'. The page features two white cards on a light gray background. The top card, titled 'Irrigation Recommendation', displays the input data: 'Crop: Rice', 'Location: meerut', and 'Soil Moisture: 10%'. It also shows a recommendation: 'Soil is very dry. Irrigate immediately for about 40–45 minutes.' The bottom card, titled 'Benefits', lists five items, each with a small icon and a green bar background: 'Save Water' (water drop icon), 'Improve Crop Health' (plant icon), 'Increase Crop Yield' (bar chart icon), 'Reduce Irrigation Cost' (flame icon), and 'Smart Weather-Based Decisions' (cloud icon).

Irrigation Recommendation

Crop: Rice
Location: meerut
Soil Moisture: 10%
Recommendation: ● Soil is very dry. Irrigate immediately for about 40–45 minutes.

Benefits

- Save Water
- Improve Crop Health
- Increase Crop Yield
- Reduce Irrigation Cost
- Smart Weather-Based Decisions

Task 4 Completed.

